

4.2 Inference using normal and t-distributions

Further Probability and Statistics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A confidence interval estimates a population parameter with stated coverage procedure. For known σ and appropriate normal conditions use z ; with unknown σ and small normal sample use Student's t with $n-1$ degrees of freedom. State null/alternative, assumptions, test statistic and conclusion in context.

Worked example: follow the reasoning

For sample mean 10, $s=2$, $n=25$, standard error estimate is $2/\sqrt{25}=0.4$; a t interval uses 24 degrees of freedom.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $n=16$, sample $s=4$ and mean $=50$, calculate standard error and degrees of freedom for a t interval. [2]

Final answer: _____

Answers and explanation

Question 1 [2 marks]

$s/\sqrt{n}=4/4=1$ [1]; degrees of freedom 15 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.